Class 10 Math - Chapter 11: Areas Related to Circles

(Note: Depending on syllabus updates, this may be termed Chapter 12. We cover the core formulas for sectors and segments here.)

🌟 Introduction

You already know that the precise Area of a circle is πr^2 and its full Perimeter (Circumference) is $2\pi r$.

In this chapter, we mathematically carve the circle into smaller distinct pieces—specifically slice-like **Sectors** and chord-divided **Segments**—and calculate their precise individual properties.

📌 Sectors of a Circle

A **Sector** is the region totally enclosed strictly by two separate radii and the corresponding arc connecting them. It geometrically looks like a slice of pizza!

- **Minor Sector:** The smaller slice (subtending angle $\theta < 180^\circ$).
- **Major Sector:** The larger remaining slice (subtending angle $> 180^\circ$).

Formulas for a Sector

Given a circle of radius r and sector angle θ precisely at the center:

1. **Length of an Arc (l):** $$ l = \frac{\theta}{360^\circ} \times 2\pi r $$

2. **Area of a Sector:** $$ \text{Area} = \frac{\theta}{360^\circ} \times \pi r^2 $$

> [!NOTE] The arc length purely scales the circumference, and the sector area purely scales the total area, mathematically proportional directly to the central angle!

⚡ Segments of a Circle

A **Segment** is the region rigorously bounded by a straight chord and its corresponding sweeping arc.

- **Minor Segment:** Bound by the sweeping minor arc and chord.
- **Major Segment:** Bound by the sweeping major arc and chord.

Formula for Area of a Minor Segment

To correctly find the area of a segment, you mathematically subtract the area of the straight triangle formed by the radii and the chord directly from the area of the sweeping sector:

$$ \text{Area of Segment} = \text{Area of corresponding Sector} - \text{Area of } \triangle OAB $$

> [!TIP] The area of the internal $\triangle OAB$ can cleanly be found using standard geometry rules, or universally using the advanced formula: $\frac{1}{2} r^2 \sin \theta$.

❌ Common Mistakes

> [!WARNING]
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> 1. **Mixing up Sector and Segment:** A sector resembles a pizza slice originating directly from the center. A segment resembles a bitten-off piece separated completely by a chord wall.

> 2. **Major Area Assumptions:** Do not simply guess the major sector area. Calculate the minor sector's exact area, and subtract it definitively from the full circle's total area (πr^2).
> 3. **Forgetting π values:** Unless stated otherwise strictly in the problem, strictly use $\pi = 22/7$. If the problem specifies $\pi = 3.14$, you MUST visually use 3.14 or lose precision points!

📝 Quick Revision

Circular Path Properties

- Geometrically, the direct distance universally covered by a rotating circular wheel in ONE pure revolution is exactly equal to its mathematical **circumference**.
- $\text{Number of revolutions} = \frac{\text{Total Distance Covered}}{\text{Circumference}}$

Summary

Areas properly related to circles test your mathematical ability to subtract geometric regions. By viewing complex shaded designs strictly as a combination of simple sectors, triangles, and segments subtracted dynamically from larger squares or parent circles, you can systematically calculate the area of any composite shape!